

Factory Reallocation & Shipping Optimization

Interactive Recommendation & Analytics Dashboard Using Machine Learning

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Abstract

This project presents a data-driven approach for analyzing factory allocation and shipping performance using order-level operational data. The system combines data preparation, exploratory analysis, machine learning, and a recommendation layer to identify orders that may benefit from factory reallocation. A Streamlit dashboard provides interactive analysis of sales, gross profit, orders, units, lead time, shipping speed, factory recommendations, expected profit, and current-versus-suggested factory allocation. The final dashboard analyzes 8,549 orders and identifies 4,608 orders for reallocation while 3,941 orders are recommended to remain at their current factory. Three regression models were evaluated for lead-time prediction: Linear Regression, Random Forest, and Gradient Boosting.

Keywords

Factory Allocation, Shipping Optimization, Machine Learning, Lead Time, Recommendation System, Streamlit, Data Analytics

1. Introduction

Factory allocation affects shipping performance, lead time, operational efficiency, and profitability. A static assignment strategy may not always provide the most suitable factory for every order. This project analyzes order and factory information to support data-driven allocation decisions. The developed system combines analytical visualizations with a recommendation workflow so that current factory assignments can be compared with suggested alternatives.

2. Problem Statement

Static factory assignments can lead to inefficient allocation decisions, increased shipping-related effort, longer lead times, and potential margin impact. The objective is to analyze the available operational data and identify orders for which an alternative factory allocation may be preferable.

3. Objectives

- Analyze factory and order performance.
- Identify potentially inefficient factory allocations.
- Compare current and suggested factory assignments.
- Analyze shipping speed and lead-time patterns.
- Estimate expected profit associated with suggested factories.
- Classify orders into keep-current and reallocation recommendations.
- Provide an interactive analytics and recommendation dashboard.

4. Dataset and Data Preparation

The project uses an order-level dataset containing operational and business attributes such as factory, product, region, division, sales, units, gross profit, lead time, shipping speed, recommendation status, risk level, suggested factory, expected profit, and predicted lead time. The dataset was cleaned and transformed before analysis and

modeling. The final dashboard contains 8,549 analyzed orders.

5. Methodology

- Data Collection and preparation
- Data Cleaning
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Machine Learning
- Factory Recommendation
- Expected Profit Analysis
- Interactive Streamlit Dashboard Development

6. Machine Learning Models

Multiple regression models were evaluated for lead-time prediction. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 .

Model	MAE	RMSE	R^2
Linear Regression	181.06	181.97	0.535
Random Forest	180.97	184.19	0.524
Gradient Boosting	181.00	182.60	0.532

7. Recommendation System

The recommendation layer classifies orders into two primary outcomes: Keep Current Factory and Reallocate. The dashboard also analyzes suggested factory assignments, expected profit, current-versus-suggested factory movement, and predicted lead time. The final analysis identified 4,608 orders recommended for reallocation and 3,941 orders recommended to remain at the current factory.

8. Dashboard Development

The interactive dashboard was developed using Streamlit and Plotly. It provides filters for Factory, Region, Division, Recommendation, Risk Level, Lead Time Category, and Shipping Speed. The dashboard includes KPI cards and analytical views covering sales by factory, gross profit by factory, sales by region and division, recommendation status, suggested factory orders, expected profit, current-versus-suggested factory comparison, predicted lead time, and monthly sales trend.

9. Results and Discussion

Metric	Observed Result
Total Orders	8,549
Total Sales	\$119,510
Gross Profit	\$78,475
Total Units	32,388

Average Lead Time	1321.4 days
Profit Margin	65.7%
Orders Recommended for Reallocation	4,608
Orders to Keep at Current Factory	3,941
Highest Expected Profit Category	Alternative Factory — \$46,239

The dashboard shows that Lot's O' Nuts is the leading factory by sales in the analyzed view, while Alternative Factory has the highest expected profit among suggested factories. The recommendation analysis provides a structured way to identify potential allocation changes rather than relying only on static factory assignments.

10. Technologies Used

Python, Pandas, Streamlit, Plotly, Scikit-learn, Microsoft Excel, and Machine Learning.

11. Conclusion

The Factory Reallocation & Shipping Optimization project demonstrates how operational data, machine learning, and interactive analytics can be combined to support factory allocation decisions. The system provides both descriptive analytics and recommendation-oriented insights, allowing users to examine performance and potential reallocation opportunities through an interactive dashboard.

12. Future Scope

- Incorporate additional real-time shipping and transportation information.
- Evaluate additional predictive and optimization algorithms.
- Integrate live operational data sources.
- Add scenario-based optimization for cost, lead time, and profit trade-offs.
- Deploy the dashboard for broader business-user access.

13. References

1. Scikit-learn documentation — machine learning model evaluation and regression tools.
2. Streamlit documentation — interactive Python application development.
3. Plotly documentation — interactive data visualization.
4. Pandas documentation — data manipulation and analysis.